Using brainless organisms to optimize complex transportation networks - a study

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Introduction

The global superpower, The United States, was built by rail. Yet Poland's rail is dated and in a poor state. In order for Poland to be able to compete with the superpowers sharing its borders like Germany or Russia it needs to be able to transport tonnes of materials quickly and cost-effectively across the nation. Unfortunately, optimizing networks is quite the difficult task. So difficult in fact, that even the most powerful modern-day computers are unable to create an algorithm which finds the optimum answer for larger instances (shubhamyadav4, 2023). Such problems, dubbed NP-hard, require us to overcome their challenges because of their answer's practical implications. The Steiner Tree Problem is such an example of an NP-hard problem that I will be solving in this paper. Basically, the question is what is the optimum network to connect a subset of vertices together. The reason why this problem is so difficult is that because we are using a subset of vertices, we can always form more vertices which could be more efficient over a plane. In the example of rail networks, we have cities (terminal vertices) which must be connected as well as junctions of railroads (nonterminal vertices) which can occur anywhere on the map (Wu, Chao 2004). Since we can create as many nonterminal vertices as we want (each with a different cost) it will not be feasible to solve a minimum steiner tree problem for large instances (many terminal vertices/large plane).

Figure 1. Terminal vertices shown as larger cities of Poland. The network can be connected not only through the terminal vertices but also through non-terminal vertices which can be located anywhere on the plane. (Medianauka.pl)

Since Japan is the leading country in railway efficiency (Statista, 2019) I decided to do some more research on how their rail network was planned and built. It was during this reading that I came upon a study conducted by the Japanese and English where a brainless slime mold replicated the optimal network conjured up by the best Japanese engineers in a matter of days (Wired, 2010). This intrigued me so I began to look for repetitions of this model to evaluate its reliability and utility.

After a deeper inquiry I came across a scientist who seemed to specialize in using slime mold as a tool to assess existing rail and highway networks of various countries. Andrew Adamatzky studied the slime molds response to many environments such as the US, Germany, Australia and even Roman roads with great success. Proving the reliability of his method and the capability of the slime mold to find solutions to complex Steiner Tree Problems. Since Polish rail has been built upon over generations (Falkowski, 2014) instead of being built all at once this logically could have led to the creation of unoptimized links in the grand scope of the modern network. Given the importance of freight transportation for the economy and sustaining efficient infrastructure, my Internal Assessment aims to create a concept of an ideal version of Poland's freight train line network and contrast a theoretical optimal network with the existing rail system. Thus, the research question is: "How does an optimum freight train line network for Poland, modeled using *Physarum polycephalum* inoculated for seven days, compare with the current Polish railway infrastructure in terms of cost and route efficiency?" This research question not only sheds light on the potential for a more optimized rail system but also helps show the broader implications of utilizing biological processes for overcoming engineering obstacles.

Background information

Slime molds belong to the Protist kingdom. They present unique behaviors and life cycles, which enable them to solve complex problems despite lacking any nervous system.

Their life cycle can be divided into several stages. Primarily they exist as multinucleate unicellular organisms (Briggs, n.d.) called plasmodium. In this stage they can move and exhibit intelligent behaviors, such as solving mazes (Nakagaki, 2000). When their environment becomes unfavorable, they can form a sclerotium, a dormant stage that allows them to survive until conditions improve. When that happens, they can become active again or reproduce through sporulation, releasing spores that can grow into new plasmodial cells (Briggs, n.d.).

SMs utilize two characteristics to forage for food: cytoplasmic streaming and pseudopodia (temporary projections of the cell membrane). Cytoplasmic streaming is the movement of the cytoplasm in a cell which allows for the transport of nutrients and enzymes along with other proteins. Driven by contractions of actomyosin complexes in the cell (Briggs, n.d.) utilizing actin filaments as a road along which motor proteins can travel by using ATP. Moreover slime molds rearrange their cytoskeleton to produce pseudopodia to navigate their environment. This feature allows the cell to engulf food through phagocytosis.

Looking into the biochemistry of slime molds, they exhibit decision-making processes that are based on a complex system of chemical reactions (Boussard, 2021). These reactions are sensitive to cues from their environment such as food allocation and light. Furthermore slime molds seem to be able to adapt to periodic trends through synchronizing their internal clock of cytoplasmic streaming oscillations with periodic trends. These abilities allow the slime mold to process information about its environment and adjust its growth and movement accordingly.

The most exciting aspect of slime mold behavior is their ability to solve NP-hard problems (computational problems that are extremely difficult to solve with algorithms, especially as they scale up). Since slime molds utilize the physics of their own structure and their environment they are free from the constraints of "data processing" bestowed upon even the most powerful modern computers. Instead, they utilize the process of adaptive network formation (Boussard, 2021) by having their protoplasm flow to areas with high concentrations of attractants (like food sources) and away from repellents (like salt).

Development of RQ

Whilst researching topics for my mathematics internal assessment, I came upon the Steiner Tree Problem. The task is to find an optimal network in a graph with vertices connected with weighted edges. Unfortunately, as more vertices are added to the graph the amount of possible networks grows exponentially - quickly getting out of reach of modern day computers. For this reason I disregarded using this topic for my math IA and kept on looking.

For my biology IA I knew that I wanted to research something that could act as a brick to help my home country, Poland. The United States is famous for being built on the railway system so I started looking for ways to modernize the Polish rail network, which brought me back to the Steiner Tree Problem. It wasn't until I saw a youtube video about a fungi-like organism mimicking the japanese intercity network, being one of the best in the world.

This led me to research how the slime mold was used to optimize other networks. I came across the Physarum being used for the US interstate highway, for the English highways and even for German highways. This turned out to be a feasible topic to investigate whilst also being able to discover interesting results.

Next I had to decide what I would measure. At first I thought to mimic previous experiments and measure the network's efficiency and mean time to get from different points. However, I was unable to grasp the methodology of how to translate the data I would obtain into those numbers. So instead, I went with comparing the cost of the railroad (which could be measured in google maps) and the paths themselves between cities of the existing rail system and the one created by the slime mold.

Preliminary study

Whilst planning my preliminary study I didn't really have a grasp on what conditions the Physarum polycephalum thrived in. All I knew was that it needed a humid environment, similarly to fungi or mold and oatmeal was a suitable food source for the organism. I prepared a petri dish with gelatine (my diy Agar replacement) mixing 3 spoons of the powder with 500ml of boiling water. I then printed out a map, to scale to fit inside the petri dish, of the largest Polish and neighboring cities connected by rail. Placing it under the now cooled and hardened gel I picked out pieces of oatmeal in correlation to city size and placed them on the petri dish. (see Figure 2) Next I proceeded to prepare the place of the experiment. I decided on conducting it in my basement as this would allow me to control light levels, temperature and humidity fairly easily. I prepared a rather sketchy camera setup for a timelapse and a light that I would keep running for the entire duration of the experiment. When the environment was ready, I placed the petri dish on a black backdrop and finally introduced the slime mold into the environment by placing it on our capitol Warsaw and waking it up with 5 drops of water using a straw and my thumb as a substitute for a pipette.

I was only able to conduct the experiment for a total of 4 days, as I was leaving home unexpectedly on a trip. This ended up not being an issue as on the last day I came upon a quite unexpected site. The previous day the slime mold looked quite healthy, although it was spreading slower than I anticipated. The next day however, the organism had hardened indicating a hostile environment, foreign mold and even a spider could be observed. I had kept the lid of the petri dish off as water kept on condensing, ruining the view on the experiment. Unfortunately this resulted in two things: first this meant that all of the water would evaporate, which I would need to replace daily but even worse this meant that foreign organisms had easy access to an environment filled with nutrients which would then start to compete with the slime mold, disrupting my experiment. Further research into optimal conditions for Physarum polycephalum showed that the organism was sensitive to light and preferred darker places. This is probably the reason why the slime mold grew slower than expected.

Figure 2. Oatmeal flakes picked in correlation to city size placed on the medium with a freshly inoculated slime mold

Now with an idea of what I was up against, I started planning an experiment that would be unfazed by such disruptions whilst still maintaining the ability to photograph the growth and inhibition of the slime mold.

Hypothesis

The ancient Polish railway infrastructure, having been developed incrementally over time, has been shown to be dated and potentially suboptimal due to its incremental construction as opposed to a uniform design (Falkowski, 2014). Modern algorithms inspired from mother nature, such as the network-forming capabilities of slime mold, offers an alternative approach to optimizing networks. *Physarum polycephalum* naturally forms efficient paths between food sources, literally bringing to life the foundations of network design without a nervous system (Werner, 2018). This process of decision making has been adapted to create optimal networks in computational models (Kay, 2022), highlighting the potential of biology to inform and improve man-made networks. In the context of rail systems, where cost-efficiency is paramount, it would be highly useful to leverage such biologically inspired designs to gain a foothold into reimagined transport routes (Shashwat, 2023). Based on this information, my hypothesis states that a network derived from the behavior of slime mold would not only be more cost-effective, evaluated by hypothetical rail costs, but would also offer more strategic routes between cities in Poland when compared to the current railway network.

Measurements

- Total length of the network (km ± 100 km): measured in ImageJ.
- Network cost (bln PLN): estimated based off of the network length.
- Suggested routes: specific pathways proposed by the slime mold model between cities

Comparative Analysis

- Transportation Network Authorship: I will compare the two types of networks, slime mold-made and man-made
- Assessed Metrics: (for each network type) assess and compare length, cost and the routes of the transportation network

Method

The investigation was conducted in order to compare the present rail network in Poland with a biologically optimized one by the Physarum polycephalum. The comparison was done by firstly introducing the slime mold to a scaled map of Poland and its cities, simulated by pieces of oatmeal similar in size to the cities. Then the slime mold was allowed to explore its new environment, connecting with every “city” on the map. After about 3 days, the slime mold already starts splicing and refining previous connections which are deemed nonoptimal in transporting nutrients across the organism. By itself the Physarum polycephalum finds the optimal network after just a couple of days, an impossible task for modern day computers, and the connections are modeled onto a computer and compared to the rail network.

Apparatus and materials

Physarum polycephalum schlerotia	1 cup 200ml	Teaspoon (5mg)	ImageJ program (±100km)

30 grams of rolled oats (Producent: “Melvit”)	1 litre of tap water	Kitchen Stove	Geogebra program
1 pair of scissors	Petri dish, r=5cm (Producent: “Germanos”)	map of Poland (1:8,000,000)	distance.to website (±0.1km)
1 set of tweezers —--------------- 1 refrigerator (Producent: “Sony”)	15 grams of universal gelatine powder (Producent: “Dr.Oetker”)	Sony a58 camera with an 18-55mm lens	glass bowl
1 plastic straw 10ml	1 litre stainless steel kitchen pot	Thermometer in Celsius (-20°C, 60°C)	soap
Aluminum tray 2.7l, 30x18x5cm	Single-use nitrile gloves (Producent: “Nitrylex”)	Eye protection goggles HF-105 (Producent: “Vorel”)	1 sharpie —----------------------- Digital kitchen scale (± 0.1g)

Table 1. Showing the apparatus and materials used during the experiment.

Ethical, safety and environmental considerations

The experiment is prepared with the use of boiling water As such, proper safety measures should be taken into consideration to avoid any incidents. Regarding the microorganism itself, it should be treated as a pathogen. Meaning, proper sterilization procedures: washing hands constantly, taking care of what you touch with “infected” hands (especially on the face area), wearing safety gloves and sterilizing your workplace. The experiment should be disposed of by adding kitchen soap to the Petri dishes after which they can be disposed of normally. In case of contamination by foreign bodies, spray with alcohol, let the alcohol sit for 3 minutes, rinse with soap and dispose of normally.

Observational Controls

Control variable	Reasoning	Method of controlling
1. growth medium	Different mediums contain different concentrations of nutrients which could affect growth of the model.	I created the gelatine jelly from one batch, causing all of the mediums for the trials to be identical.
2. humidity	The slime model prefers moist conditions. If humidity would differ between trials this could significantly affect growth.	I set the trials on a tray of water which through evaporation would maintain equally high humidity for all of the trials.
3. temperature	Different temperatures could affect the behavior of the slime mold.	All of the trials were done in the same location with a thermometer to make sure a stable 20°C was maintained.
4. light	Slime molds prefer dark conditions. Different light conditions between trials would affect growth.	The trials took place in a room with no access to light except during photographing progress.
5. sterility	We want to avoid any competition between species during the experiment as this would change the behavior of the model.	Proper sterile conditions were kept during the experiment, such as wearing clean gloves, boiling the medium, not tilting the lid during opening, etc.
6. map scale	This is needed to produce reliable data between trials so the distances between cities for the model are the same.	One map template was printed and used for all trials.

7. initial placement of oatmeal flakes	Vital for making sure the slime mold interacts with the actual locations of cities from the map.	The flakes were precisely placed in their place using the map template.
8. Oatmeal flake size	This ensures that the slime mold considers the size and thus importance of cities during its growth.	The oatmeal flakes were chosen by size for the respective cities so that larger cities had larger flakes.

Table 2. Showing the control variable as well as the reasoning for and the method of controlling.

Procedure

1. Preparation of Tools and Materials
    a. Gather all necessary equipment: petri dishes, gelatine powder, water, spoon,, map of Poland, oatmeal flakes, tweezers, tray, dark location, camera, software, soap, etc.
2. Petri Dish Setup
    a. Boil 500ml of water and dissolve 15 g of gelatine powder into it to create a 3% gelatine concentrated jelly
    b. Pour a 3mm layer of the gelatine solution into each of the 5 petri dishes.
    c. Cool the dishes in a refrigerator for two hours to solidify the gelatine.
3. Map and Oatmeal Flake Preparation
    a. Print a scaled map of Poland.
    b. Prepare oatmeal flakes to correspond with the cities on the map.
4. Assembling the Experimental Model
    a. Position the solidified gelatine petri dishes over the printed map.
    b. Using tweezers, place oatmeal flakes on the gelatine at their respective city locations on the map.
5. Environment Setup for Physarum
    a. Identify a cool, dark space for the experiment.
    b. Fill a tray with water, place it on a flat surface, and center the petri dish on it.
6. Introducing Physarum polycephalum
    a. Put on your gloves and protective glasses and find a well-ventilated area.
    b. Place the Physarum polycephalum on the capital city's oatmeal flake using tweezers.
    c. Add a few drops of water onto the Physarum using a straw as a pipette.
7. Repeat steps 3-6 four more times for a total of five trials
8. Maintaining Experimental Conditions
    a. Cover the petri dishes and tray with a glass bowl to maintain high humidity.
9. Daily Observation and Data Collection
    a. Take daily photographs of the petri dishes for one week.
10. Quantitative Analysis
    a. On the 7th day, analyze the photograph using ImageJ software to trace the Physarum network and collect numerical data.
11. Post-Experiment Cleanup
    a. After completion, spray the petri dish with soap.
    b. Dispose of the experimental materials safely.

Data analysis

Quantitative

Transport network length of the *Physarum polycephalum* model to real world scale (km ± 100)				
Repetitions				
1	2	3	4	5
3465	3121	2857	3311	3465

Table 3. Quantitative raw data measured in imageJ and distance.to showing the length in km of the theoretically optimal railway networks of 5 different Physarum polycephalum trials inoculated for 7 days.

Results from the raw data (Table 3) were used to calculate the mean of the length of the simulated railway network created by the five slime molds (Table 4). In addition, the cost (averaged at 30 mln PLN / km (Wyry, n.d.)) and standard deviation of the resulting networks were calculated and were compared to the actual rail network present today (Ministerstwo Infrastruktury, n.d.).

	Slime mold-made network		Man-made network	
	Length of network (km ± 100)	Cost of network (bln PLN)	Length of network (km ± 100)	Cost of network (bln PLN)
Mean	3243.8	97.314	3838.0	115.140
SD	258.3934984	7.751804951	-	-
Median	3311	99.33	3838.0	115.140
Range	608	18.24	0	0

Table 4. Quantitative processed data showing the contrast between the mean (from 5 repetitions) cost and length of the theoretically optimal rail line network modeled using Physarum polycephalum inoculated for 7 days and the actual rail system along with the models' standard deviation.

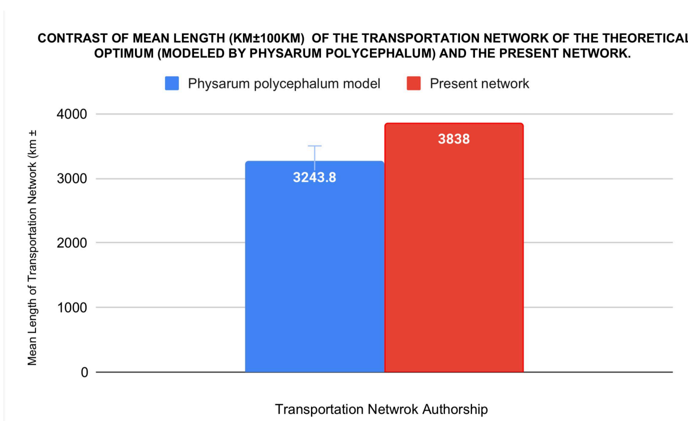


Chart 1. Bar graph visualizing the contrast of the mean length (km ± 100) of the transportation network of the theoretical optimum (modeled by Physarum polycephalum) and the present network

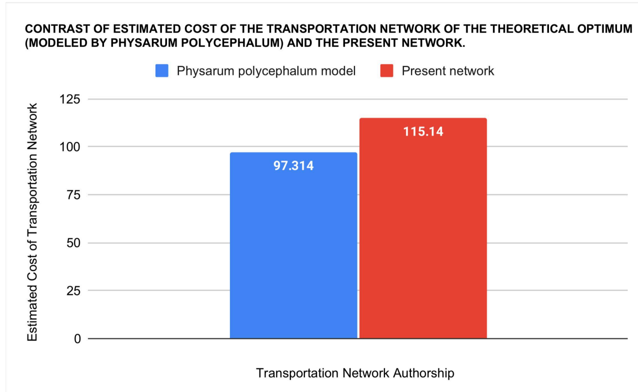


Chart 2. Bar graph visualizing the contrast of the estimated cost of the transportation network of the theoretical optimum (modeled by Physarum polycephalum) and the present network

My hypothesis was: “a network derived from the behavior of slime mold would not only be more cost-effective, evaluated by hypothetical rail costs, but would also offer more strategic routes between cities in Poland when compared to the current railway network”. According to the results in Table 4 and Charts 1,2 my hypothesis correctly predicted the results. The *Physarum polycephalum* was successfully able to find a more optimal network for transport.

The data presented in Chart 1 is to compare the two sets, so a bar graph was chosen. The standard deviation of the simulated network is just under 10% of the mean which shows that the model still has room for improvement. However, even the “least” optimal network presented by the *Physarum* is still over 300 km shorter and 10.5 billion PLN more cost effective than the present network. The data for the length and subsequent cost of the present network was taken from the Polish government website (Ministerstwo Infrastruktury, n.d.), so it has no standard deviation.

Qualitative

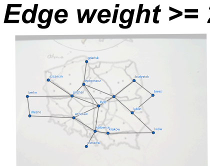
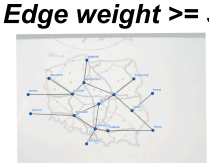
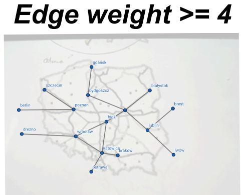
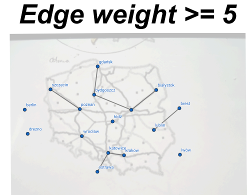
Edge weight >= 2 	***Edge weight >= 3*** 
Edge weight >= 4 	***Edge weight >= 5*** 

Table 5. A compiled network from the 5 trials, one weight was added for every instance of an edge in a network (a possible total of 5)

After measuring the length of the networks in ImageJ, I computed the slime mold-made routes into geogebra onto a background of the man-made network. Increasing the weight of each route depending on the frequency of a route occuring during the 5 trials. From this visualization we can observe a couple characteristics.

For one, all routes created by the slime mold occured at least twice. This is very interesting as it serves as direct evidence that the proposed networks are optimal as all of the routes created ended up serving some need causing them to be replicated over different trials.

Next, looking at edge weights >=3 and >=4; we can observe that the east-west routes are more important compared to north-south routes as the former detach last into two parts.

Lastly, looking at edges which occured in all trials we can observe the most vital connections between cities for the layout of Polish cities to connect the country together.

It is worth noting that the observations made are only in regards to transport economy and efficiency as that is all the slime mold model interacts with. The model has no concept of strategic layouts (eg. military strategy) which could cause some differences between the model and ten-t network.

Statistical analysis

The aim of the comparative analysis was to compare the efficiency of the slime mold-made railway network against the existing man-made railway network in Poland. Due to the nature of the collected data—where the slime mold network's data consists of five distinct trials and the existing network represents a single data point—a traditional statistical test like the Mann-Whitney U could not be performed. Instead, a comparative analysis was conducted.

The comparative analysis involved calculating the mean, range, median and standard deviation for the slime mold network data across the five trials and contrasting these with the single data point from the existing network. This approach provided a qualitative comparison of the observed differences in network length and cost.

In addition it should be noted that the theoretical optimal network proposed by the slime mold networks on average is 15.5% (calculated using Figure **3**) more efficient in regards to length and thus, cost.

$$\boldsymbol{C} = \frac{x_2 - x_1}{x_1}$$

Figure 3. The formula for calculating the percentage change between two variables.

Although the sample sizes and analysis methods limited the robustness of statistical conclusions, the comparative statistics provided insight into potential efficiencies of the slime mold-designed network. The comparative cost and length from the slime mold data indicated a trend towards a more efficient network, with an average reduction in both cost and length compared to the existing network.

Future studies with more balanced and larger sample sizes for both types of networks would enable the use of comparative statistical tests to confirm the trends observed in this preliminary study.

Conclusion

The investigation obtained results showing that the slime mold network was far more optimized in regards to the Polish network. Precisely 15.5% more efficient over the man-made network, including reducing the total length of the network on average by 595 km as well as saving 17.85 bln PLN in laying the tracks which can be spent elsewhere while at the same time creating a more efficient transportation network. Evidently the results support the hypothesis and despite lacking the ability to conduct a traditional statistical test we can conclude that the differences are certainly significant and profound.

These findings align with the countless studies conducted by Adamtzky-Ilachinski eg. in (Adamatzky-Ilachinski, 2012) which investigated what networks Physarum polycephalum trials created on the map of the United States and showed that the Physarum was very adept at creating transport networks by successfully imitating the transportation network of a nation that was literally built on rail. The reasoning behind exactly how such a relatively simple organism is able to solve such complex problems is still unknown to us. Today we only have a handful of theories as to how the slime mold operates (Boussard, 2021) mainly that they utilize complex chains of chemical reactions along with cytoplasmic streaming and its mechanics which have evolved to react to specific stimuli allowing the unicellular organism to grow in appropriate directions and splice unneeded / nonoptimal connections similar to neural pruning. Nonetheless, the huge difference between the theoretical optimum modeled by the slime mold and the man-made network shows that Polish modern rail is nonoptimal and could use much improvement, which just further goes to show the implications of utilizing biologically inspired solutions to our engineering problem.

Evaluation

My investigation allowed me to answer the research question and support the hypothesis created for the experiment and the *Physarum polycephalum* trials showed that a more optimal transport network is feasible for the Polish freight train system.

Strengths

The experiment has many strengths which allowed me to obtain reliable data. First off, the *Physarum polycephalum* is easy to buy online. Furthermore, once bought it is easy to create more of the slime mold. Secondly, other than the test subject, the materials and apparatus used during the experiment are all readily available in homes and can be bought in stores if need be. This allowed me to not have to organize special trips to prepare any materials/apparatus. In addition, a large surplus of gelatine concentration was synthesized which allowed for exact forming of petri dishes without me having to worry about some concentration staying in the pot. This decreased the uncertainty of the thickness of the gelatine concentration in the petri dishes. In addition, the preliminary study allowed me to find and solve problems early in the experimental process without having to worry about materials going to waste. This allowed me to save money, resources and frustration later down the road. Furthermore, a total of 5 trials were used to be able to come to a more reliable theoretically optimal network which would not be as accurate with just a single trial as even the smallest starting condition can have a massive impact on the final outcomes besides taking into consideration that the slime molds had slightly different genetic makeups as they were separate organisms. Lastly, a deep analysis was done on both the quantitative data and qualitative data which allowed me to come to thorough conclusions on the implications of the biologically designed network.

Limitations

Despite all of its strengths, the methodology had some limitations. First off, not all cities were placed into the environment of the *Physarum* as some were so close that the oatmeal flakes simulating the cities would be on top of each other. This could be resolved by using a larger-scaled map, creating a larger area for the placement of oat flakes. Another limitation would be the fact that only one man-made network was used for comparing the optimization capabilities of a slime mold model. Instead, the same amount of man-made networks (5) could have been evaluated regarding the same geographical layout with the slime mold models in order to truly test if they are better at optimization. A third limitation was the narrow focus point for the model. Being provided only with “data” of the locations of Polish cities and at most the closest neighboring countries' cities, the slime mold did not have the opportunity to create a network which spand a larger part of Europe, which is significant ever since we have joined the EU. To solve this, a much larger scale would need to be used along with larger petri dishes to be able to realistically simulate a larger portion of Europe and thus get more reliable data for optimized Polish routes.

Future suggestion for idea extension

In future research a larger scaled map would prove to be more useful for finding the optimum network for a Euro-centric transport network. Not only would this produce a network that could be used by the entire continent, but it would also further create a more precise network for the future Polish freight railway. Furthermore, it would be a good idea to use the gravitropic properties of *Physarum polycephalum* to further produce even more accurate results of a highly optimized network. Further research in this Euro-centric area could also be used to model catastrophes like war or needs of isolating regions by introducing sodium chloride crystals into specific regions (eg. nuclear reactor sites) to observe potential emergency scenarios of rerouting existing networks after catastrophic events. Increasing the safety of others by giving us the ability to foresee the work needed to be done to repair essential transport routes for example food.

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